Parametric Equations (8.4) Name ______________________

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write "2" in the blank. Solve that problem, find your answer and write "3" in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $x^2 + y^2 = 4$ # ________ Given $x = t+1,\ y = t^2$, find y when $t = 3$.	ANSWER: $x = y^2$ # ________ Given $x = t-1,\ y = 2t$, find x when $t = 5$.
ANSWER: $(2, 4)$ # ________ Given $x = \cos t,\ y = \sin t$, eliminate the parameter.	ANSWER: $(2, -2)$ # ________ Given $x = 4t,\ y = t$, find y when $x = 8$.

Answer: 4

#______ Given $x = 3t,\ y = 9t^2$, eliminate the parameter (write y in terms of x).

Answer: 9

#______ Given $x = 2t,\ y = t - 1$, eliminate the parameter (write y in terms of x).

Answer: $y = x^2$

#______ Given $x = t + 2,\ y = t - 2$, find the point when $t = 0$.

Answer: $x^2 + y^2 = 1$

#______ Given $x = t^2,\ y = t$, eliminate the parameter (write x in terms of y).

Answer: 2

#______ Given $x = 2\cos t,\ y = 2\sin t$, eliminate the parameter.

Answer: $y = \frac{x}{2} - 1$

#______ Given $x = t,\ y = 3t - 2$, find the point when $t = 2$.